

Final Report

Predictioneer

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1.Problem Statement

Ebola, like many infectious diseases, poses a significant public health risk. Early prediction of expected cases and deaths in various regions is crucial for effective resource allocation, timely intervention, and containment efforts.

In this challenge, the goal is to develop a machine learning model that can predict:

- The **number of deaths**
- The **number of confirmed cases**
- The **Case Fatality Ratio (CFR)**

The dataset includes:

- **Geographical information** (latitude & longitude)
- **Reported deaths in some regions** (while missing in others)
- **Existing CFR data** for specific locations

Using this data, the model should forecast deaths, confirmed cases, and CFR for unreported regions.

What Needs to Be Done

- Analyze spatial and epidemiological data
- Build a machine learning model to predict Ebola cases & fatalities
- Identify patterns in disease spread

- Provide actionable insights for early intervention & resource allocation

This project will contribute to **early outbreak detection, targeted resource distribution, and improved public health response strategies.**

2. Our Research

A). Transmission

Ebola Virus Disease (EVD) spreads through the following means:

1. Person-to-Person Transmission

- Direct contact with bodily fluids (blood, saliva, urine, feces, vomit, breast milk, amniotic fluid) of an infected person.
- Contact with contaminated clothing, bedding, needles, or medical equipment used by an infected individual.

2. Animal-to-Person Transmission

- Ebola can spread through contact with infected animals such as bats or non-human primates.

3. Contact with Sick or Deceased Individuals

- **Ebola can be transmitted through direct contact with a sick or deceased person infected with EVD.**

B). Case Fatality Rate (CFR)

The Case Fatality Rate (CFR) is the proportion of people who die from a specific disease among all individuals diagnosed with it.

C).Spatial Analysis in Disease Transmission

1. Spatial Lag

- Spatial lag captures the average value of a target variable (e.g., deaths) for neighboring points within a defined geographic radius.
- This feature accounts for spatial dependencies by considering how a location is influenced by its surroundings.

2. Clustering vs. Spatial Lag

- Clustering: Groups locations into distinct regions and aggregates values within those clusters (e.g., cluster mean deaths).
- Spatial Lag: Focuses on individual locations and aggregates values from specific neighboring areas within a given radius.

D).Hot Spot Analysis

- The spread of EVD is higher in temperate zones than in tropical zones.
- Drier regions tend to experience lower transmission rates.
- Reanalyzing these trends multiple times can help gain deeper insights.

E).Key Insights from the Analysis

- ✓ Temperature is a critical factor in virus transmission and survival.
- ✓ The virus thrives in cooler environments but struggles in warmer climates.
- ✓ Highest transmission rates occur in temperate zones (5–10°C).
- ✓ The tropical zone (>25°C) exhibits minimal transmission.

F).Supporting Research

Findings from Schulman and Kilbourne's and Lowen et al.'s research on influenza transmission models reinforce the impact of temperature and humidity on viral spread:

- At 20°C, transmission efficiency of influenza A/H3N2 showed a bimodal dependence on relative humidity (RH):
 - Maximal at 20–35% RH

- Poor at 50% RH
 - Moderate at 65% RH
 - Absent at 80% RH
- At 5°C, transmission was more efficient than at 20°C, with efficiency decreasing as RH increased.
- At 30°C, transmission was completely abolished, regardless of RH.

G).Key Takeaways

- ✓ Cooler temperatures (5–10°C) enhance viral transmission.
- ✓ Higher temperatures ($\geq 30^\circ\text{C}$) significantly reduce transmission.
- ✓ Humidity affects transmission, but temperature remains the dominant factor.

This information is crucial for disease prevention, outbreak forecasting, and public health interventions.

3).Analysis of Dataset :

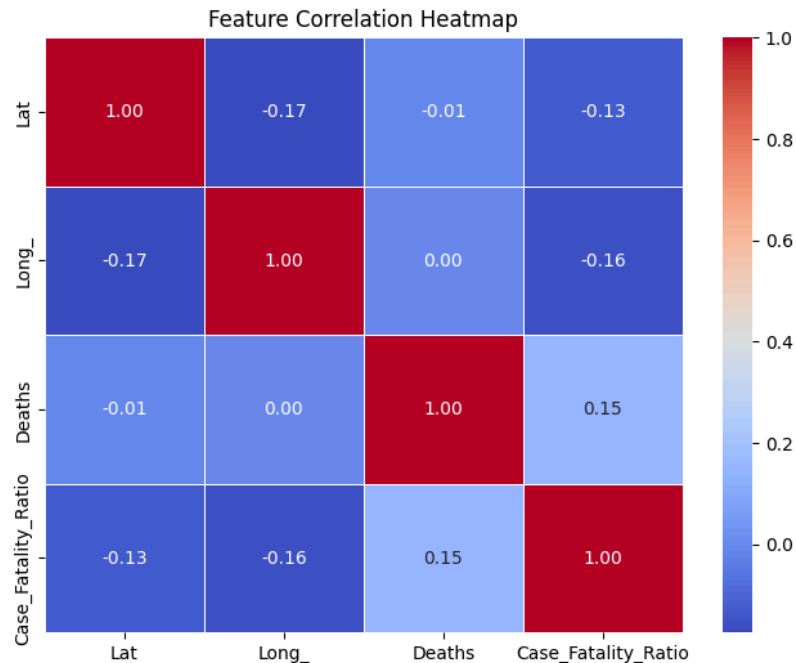
- The first thing we did was analyze the distribution of values in the dataset.

```

Data columns (total 4 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Lat                  3920 non-null   float64
1   Long_                3920 non-null   float64
2   Deaths              2375 non-null   float64
3   Case_Fatality_Ratio  3920 non-null   float64

```

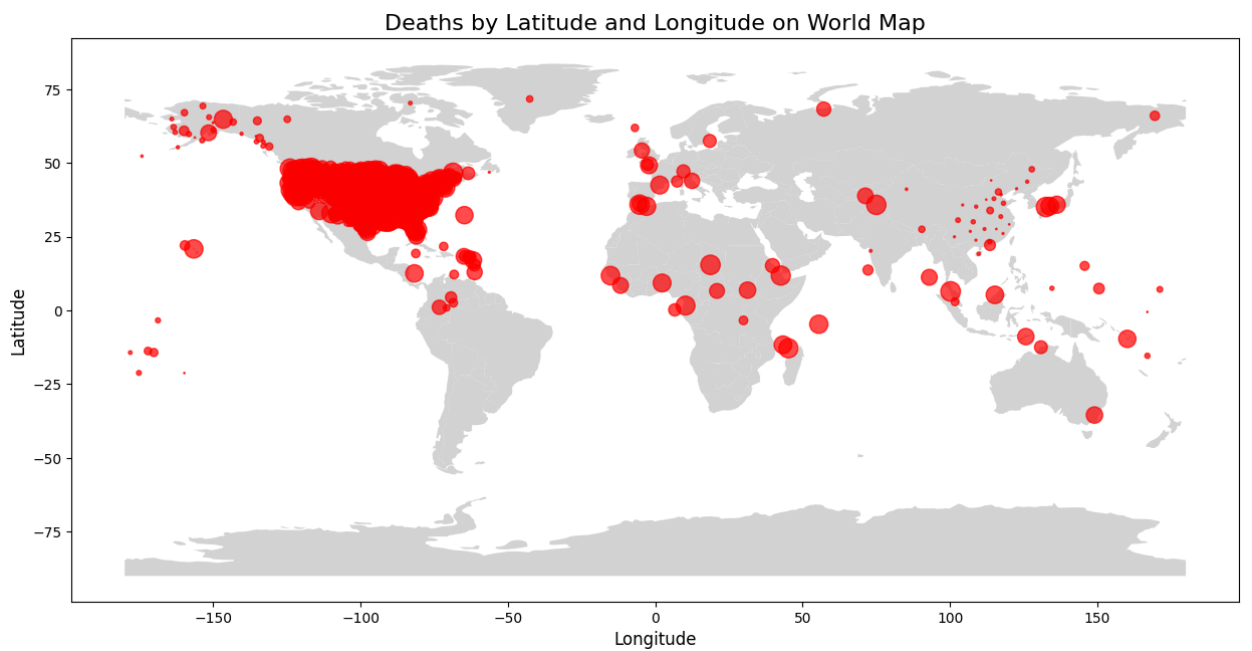
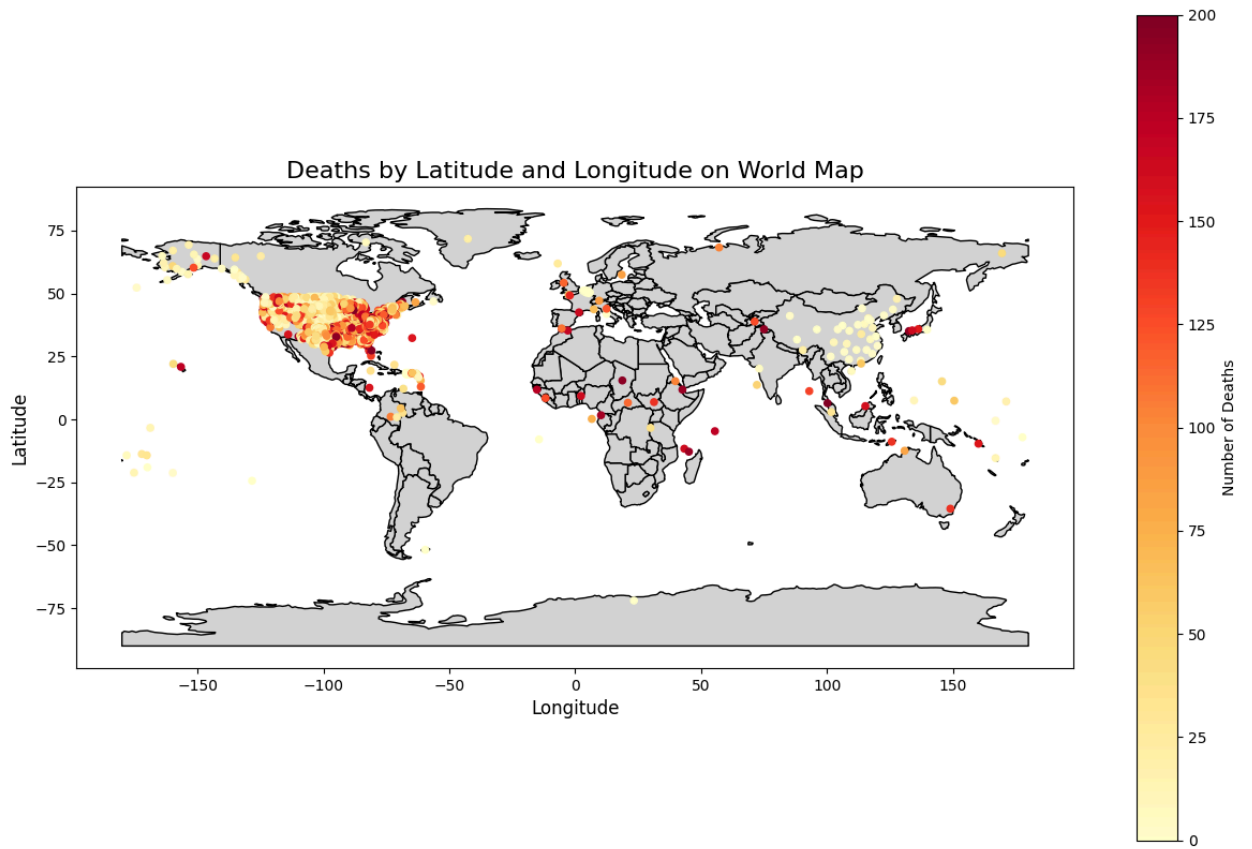
- We looked at the correlation along various features as well. There was very little correlation among features suggesting there is no linear relationship between them.
- Our target columns are **Deaths and Case_Fatality_Ratio**.



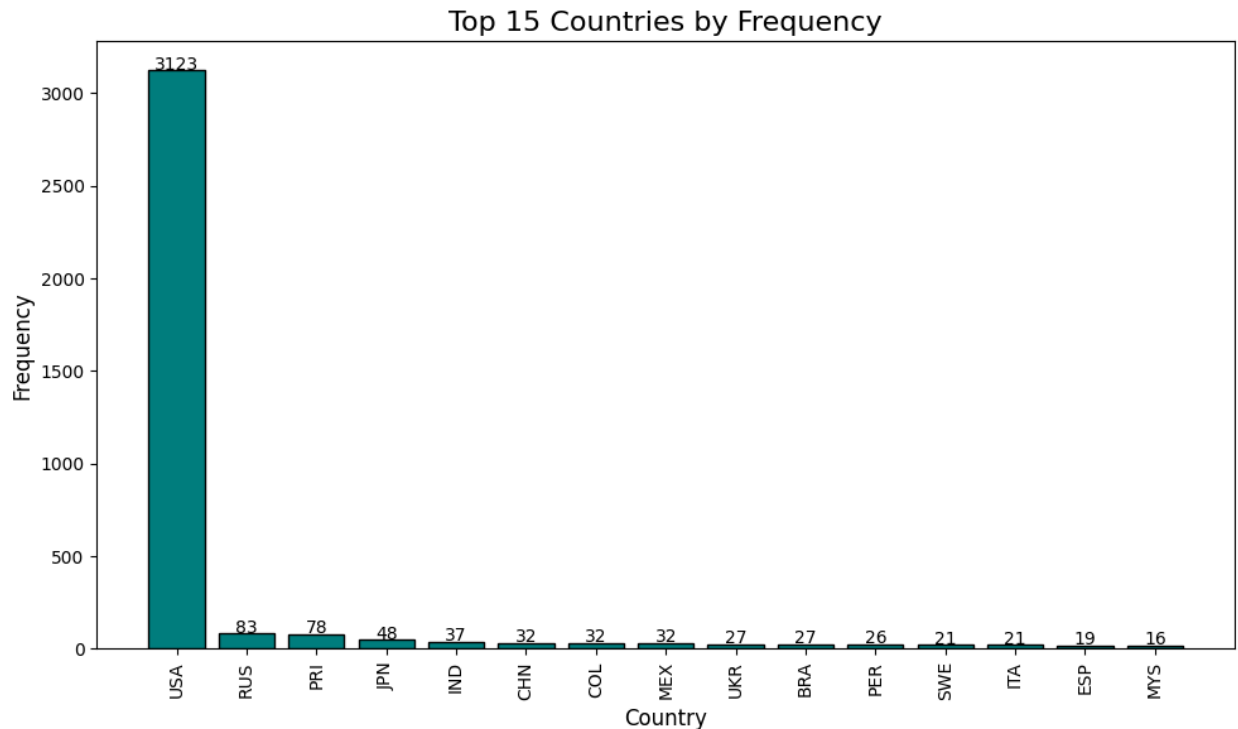
- We also looked at the distribution of death counts in different regions along with other statistics on the death count column.

4).Data Cleaning :

- We first removed all rows where Latitude and Longitude had a nan value.
- We also removed rows where CFR exceeded 100 as that is impossible.
- With the help of geopandas library, we created a **GeoDataFrame** in order to visualize the distribution of deaths all over the world.
Country data was obtained by loading a world map shapefile from **Natural Earth**.
- We also looked at the geographic distribution of null values in the deaths column along with country-wise distribution of deaths.



- Frequency Graph also show that most of the data is in USA only so this dataset is very toward to USA.



5).Imputation of Missing Values

Before moving forward we need to fill in all the null values present in the deaths column. We analyzed various different methods to do so. We have elaborated on them below.

(a) Imputation before Feature engineering

(i) IDW :

- This imputation method works under the assumption that the influence of a known data point on an unknown point decreases with distance. In order to implement this, we first separate our dataset into 2 parts, one with death values known and another with death values unknown. Then, a distance matrix is computed to calculate the distance between points. Using the **IDW function**, we apply the inverse distance weighting formula for each missing value

- The IDW function calculates a weighted sum of **all known deaths** for each unknown point, using the inverse of the distances between the points as the weights.
- We used Haversine distance to calculate the distance between 2 points.

φ_1, φ_2 are the latitude of point 1 and latitude of point 2

λ_1, λ_2 are the longitude of point 1 and longitude of point 2

$$\text{hav}(\theta) = \sin^2\left(\frac{\theta}{2}\right) = \frac{1 - \cos(\theta)}{2}$$

$$\begin{aligned} d &= 2r \arcsin\left(\sqrt{\text{hav}(\varphi_2 - \varphi_1) + (1 - \text{hav}(\varphi_1 - \varphi_2) - \text{hav}(\varphi_1 + \varphi_2)) \cdot \text{hav}(\lambda_2 - \lambda_1)}\right) \\ &= 2r \arcsin\left(\sqrt{\sin^2\left(\frac{\varphi_2 - \varphi_1}{2}\right) + \left(1 - \sin^2\left(\frac{\varphi_2 - \varphi_1}{2}\right) - \sin^2\left(\frac{\varphi_2 + \varphi_1}{2}\right)\right) \cdot \sin^2\left(\frac{\lambda_2 - \lambda_1}{2}\right)}\right) \\ &= 2r \arcsin\left(\sqrt{\sin^2\left(\frac{\varphi_2 - \varphi_1}{2}\right) + \cos \varphi_1 \cdot \cos \varphi_2 \cdot \sin^2\left(\frac{\lambda_2 - \lambda_1}{2}\right)}\right). \end{aligned}$$

- The above image shows the formulae used to calculate Haversine distance which calculates the great-circle distance between two points on a sphere (such as Earth) given their latitudes and longitudes. It accounts for the Earth's curvature, making it more accurate than Euclidean distance for geographical coordinates.

(ii) Ball Tree :

We start by separating our dataset into 2 parts, one with death values known and another with death values unknown. This imputation method is also similar to IDW but it finds the k nearest neighbours of each unknown data point based on Haversine distance. It then computes the **mean** of the death values from the **k nearest neighbors** and use it to fill in the missing values for the unknown locations.

(iii) Kriging :

- It is a geostatistical imputation method that makes use of both data and spatial correlation between points.
- It assumes that data points close to each other are more likely to have similar values.
- It makes use of a mathematical model called a variogram to describe the similarity between 2 points based on distance. Kriging predicts missing values based on the known data and provides an estimate along with a measure of uncertainty
- The variogram plots the **semivariance** (a measure of difference between data points) against the **distance** between those points. We made use of a **Gaussian** variogram model to estimate how much influence nearby points will have on each other when predicting missing values.

(b) Imputation after Feature Engineering
RandomForestImputer

6).Feature Engineering :

(a) Temperature : We found data regarding average surface temperature in each country over the last 10 years and used this as our source of temperatures for this feature.

Now why use the feature of temperature?

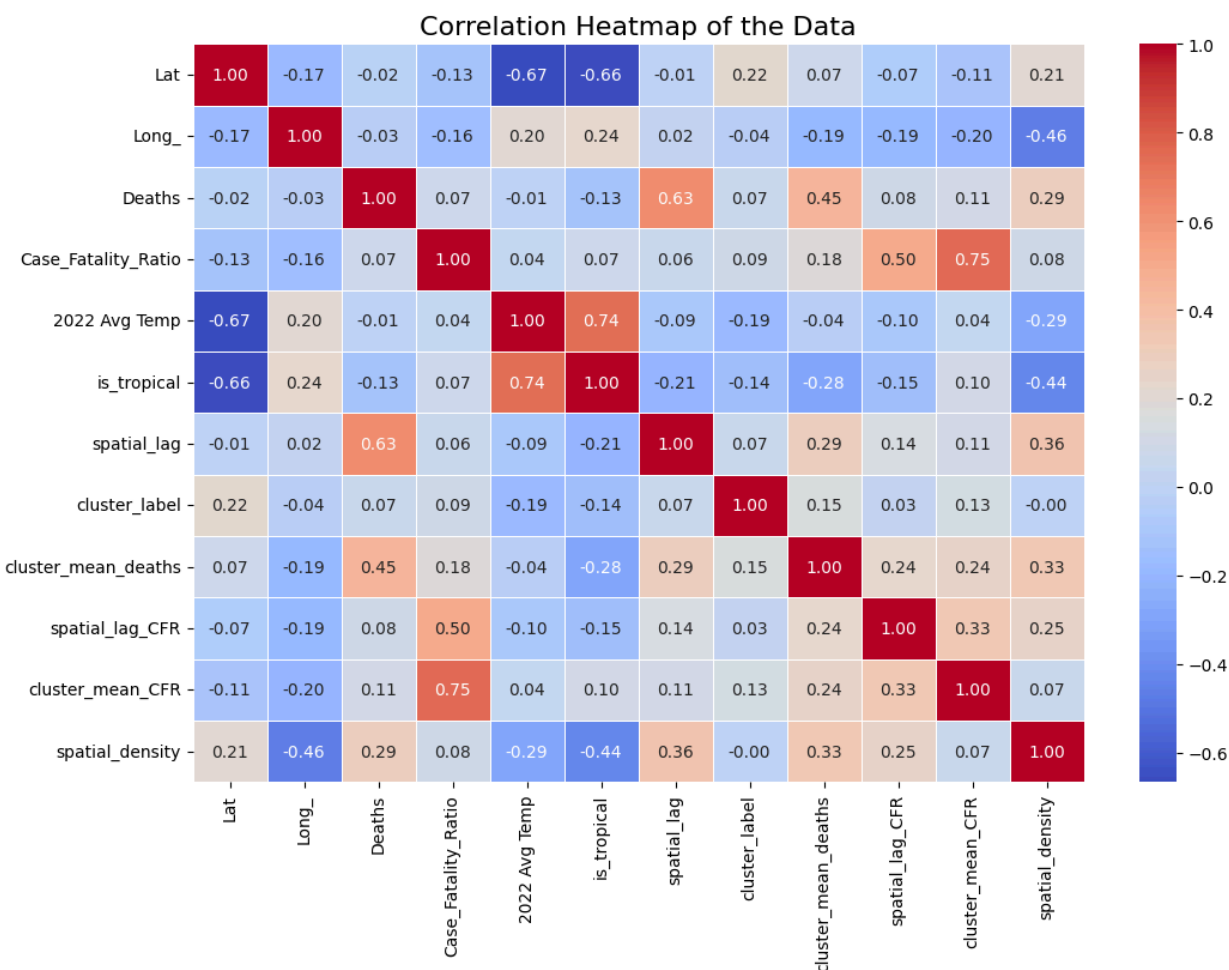
- The virus thrives most effectively in temperate zones, where the average temperature is between 5°C and 10°C. This temperature range is particularly conducive to viral survival and transmission, likely because the virus can remain stable and infectious at these lower temperatures. Consequently, the rate of transmission is highest in these regions.
- In contrast, tropical zones, where the average temperature exceeds 25°C, see minimal virus transmission. Warmer temperatures above this threshold are not ideal for the virus to remain stable, leading to reduced transmission. We found studies online supporting our argument.

(b) Spatial Features : The Spatial lag feature calculates the average value of a target variable from neighbouring locations within a particular geographic radius. We made use of a BallTree algorithm with Haversine distance to efficiently find all nearby points within the specified radius and compute their mean value. This feature reflects the idea that nearby locations tend to influence each other. For example, higher death rates in one area may be associated with higher death rates in neighboring areas due to similar environmental or socio-economic factors. Spatial lag helps account for spatial dependencies in the model enabling it to make accurate predictions based on its surrounding locations.

(c) Cluster Mean Deaths : We made use of the DBSCAN algorithm to cluster locations into distinct regions based on their proximity and then calculate the mean death rate per cluster. Locations that belong to the same cluster are assumed to have similar characteristics, which helps the model make more region-specific predictions. Points that are not part of any high-density region and don't have sufficient neighbors to form a cluster are labeled as noise. Original death counts are retained for outliers. This helps the model capture **local variations** and **regional trends**

(d) Spatial Density : This feature tells us about how concentrated data points are within a given region around a specific location. Using **cKDTree**, we calculated the number of data points within a **10 km radius**. This tells us how crowded or spread out a region is. High spatial density might indicate areas with a larger number of deaths, while low density might correspond to rural or less populated areas. This helps the model identify potential hotspots.

After making all feature we have this HeatMap



This give very crucial information about the relationship between input and target features.

7).Model Training :

(Death Model, CFR Model)

(a) Machine Learning Models 😊

We used the following models:

- RandomForest: RandomForestRegressor(n_estimators=100, random_state=42)
- XGBoost: xgb.XGBRegressor(n_estimators=100, random_state=42)
- GradientBoosting: GradientBoostingRegressor(n_estimators=100, random_state=42)
- LinearRegression: LinearRegression()
- Lasso: Lasso(alpha=0.1)
- ElasticNet: ElasticNet(alpha=0.1, l1_ratio=0.5)
- Ridge: Ridge(alpha=1.0)
- KNeighbors: KNeighborsRegressor(n_neighbors=5)
- Bagging_DecisionTree: BaggingRegressor(estimator=DecisionTreeRegressor(), n_estimators=50, random_state=42)
- Bagging_XGBoost: BaggingRegressor(estimator=xgb.XGBRegressor(n_estimators=50), n_estimators=50, random_state=42)
- Bagging_KNeighbors: BaggingRegressor(estimator=KNeighborsRegressor(n_neighbors=5), n_estimators=50, random_state=42)
- AdaBoost: AdaBoostRegressor(n_estimators=100, random_state=42)
- SVM: SVR(kernel='rbf', C=100, epsilon=0.1)

(b) Deep Learning Models

Simple Neural Network Architecture:

The **simple neural network** has a straightforward design with three dense layers. Each hidden layer is followed by batch normalization, LeakyReLU activation, and dropout for regularization. The network processes input features through a series of transformations to make predictions with a single output layer.

Advanced Neural Network Architecture:

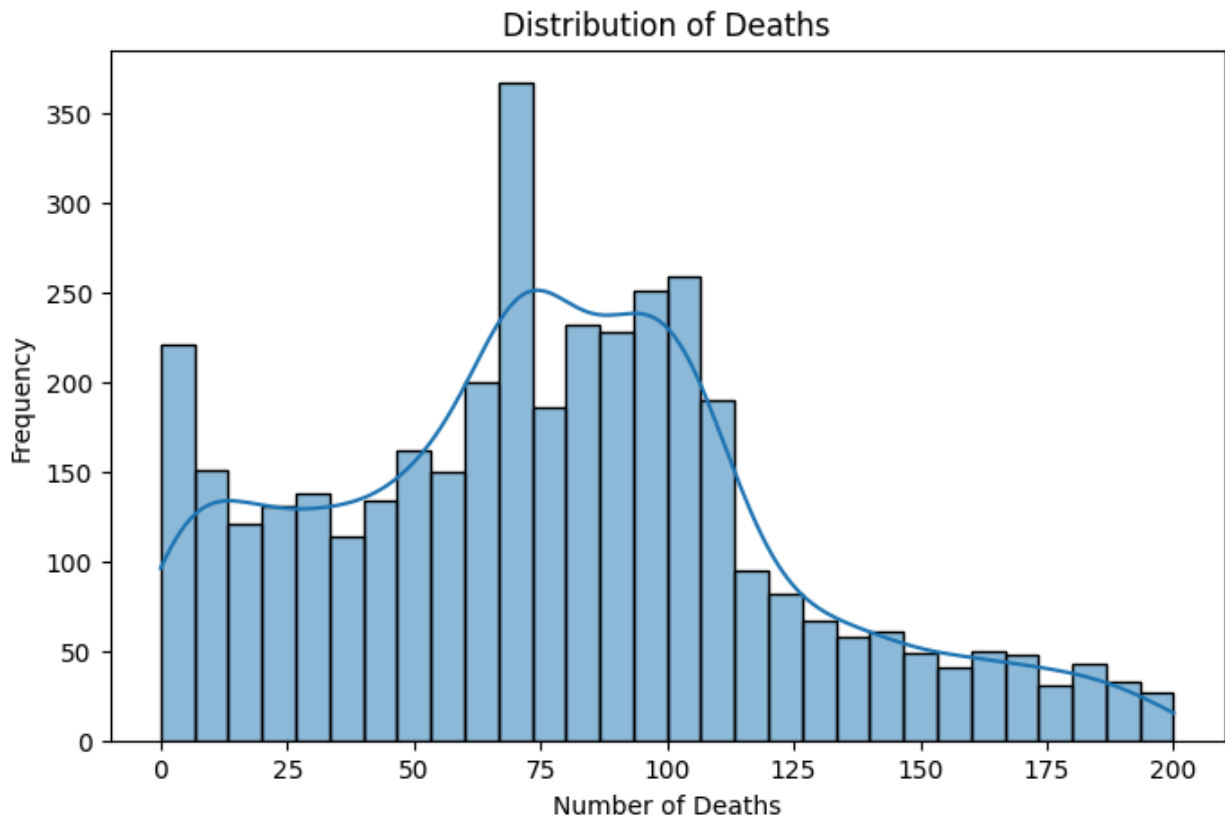
The **advanced neural network** introduces more complexity with two branches. The **deep branch** includes multiple dense layers, residual blocks with skip connections, and batch normalization. The **shallow branch** captures linear patterns using simpler layers. These branches are then concatenated, and the combined output is processed through a final set of layers with regularization for better performance.

8).Results

We imputed the death columns with 3 methods as mentioned above (IDW,Ball Tree,Kriging).

1).IDW Imputation -

This is the distribution of deaths after imputation which resembles a gaussian distribution.



Death Model

Features Used in the Analysis

The model utilizes the following key features for predicting disease transmission patterns:

1. Latitude (**Lat**) – Represents the geographical latitude of a location.
2. Longitude (**Long_**) – Represents the geographical longitude of a location.
3. 2022 Average Temperature (**2022 Avg Temp**) – The average temperature of the region in the year 2022, crucial for analyzing the effect of temperature on virus transmission.
4. Cluster Mean Deaths (**cluster_mean_deaths**) – The average number of deaths within a defined cluster, providing insights into regional mortality patterns.

5. Spatial Lag (**spatial_lag**) – Captures the average value of deaths from neighboring regions within a specified geographic radius, helping to model spatial dependencies.
6. Spatial Density (**Spatial_density**) – Represents the number of death density in a given area, influencing the likelihood of disease spread.

a) Machine Learning Models

Model	RMSE
RandomForest	33.40
XGBoost	35.32
GradientBoosting	32.47
LinearRegression	33.15
Lasso	33.16
ElasticNet	33.22
Ridge	33.15
KNeighbors	34.99
Bagging_DecisionTree	33.45
Bagging_XGBoost	32.60
Bagging_KNeighbors	34.20
AdaBoost	35.49
SVM	32.74

Best Model and Best Combinations of Features:

Best Model: Bagging_XGBoost

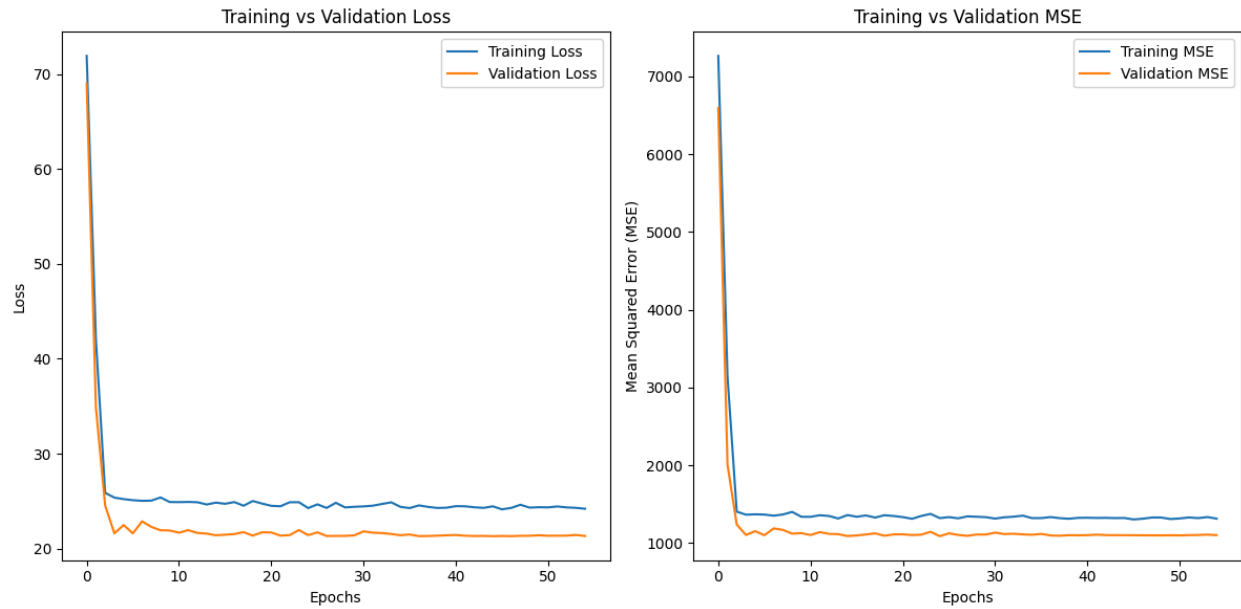
Best Feature Set: ('Lat', 'Long_', '2022 Avg Temp', 'cluster_mean_deaths')

Best RMSE: 32.23

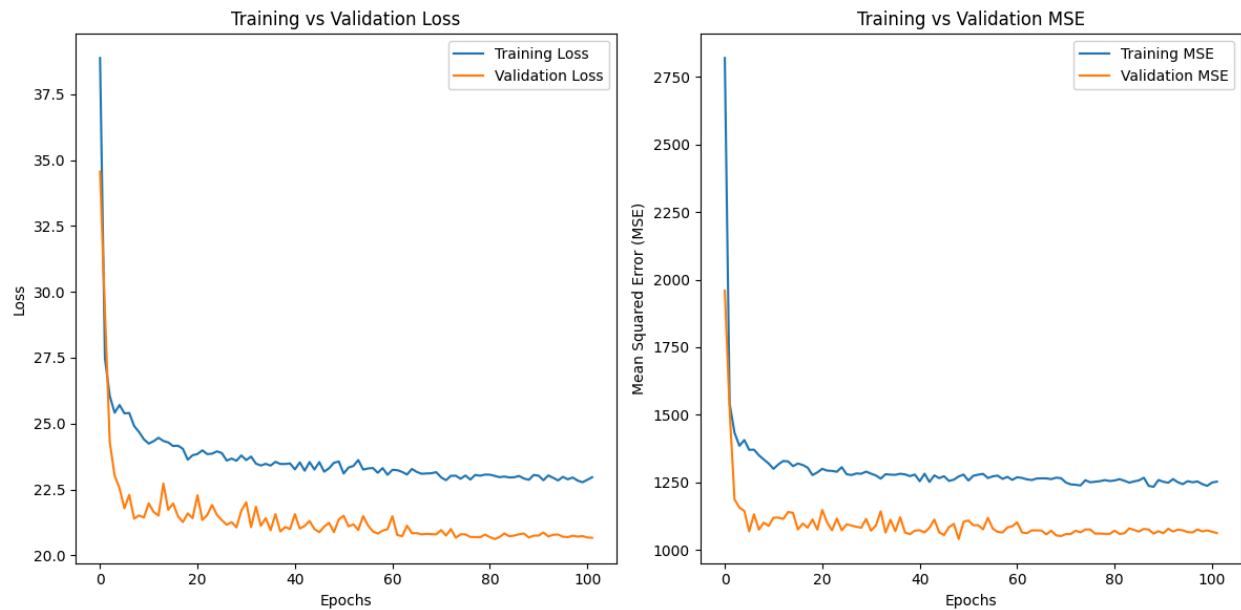
Final Model RMSE: 32.23

b) Deep Learning Models

- 1).Simple NN – : Death Prediction RMSE: 33.17



2). Advanced NN with residual Block - Death Prediction RMSE: 32.54



CFR Model

Features Used in the Analysis

The model utilizes the following key features for analyzing disease transmission and Case Fatality Rate (CFR):

1. Latitude (**Lat**) – Represents the geographical latitude of a location.
2. Longitude (**Long_**) – Represents the geographical longitude of a location.
3. 2022 Average Temperature (**2022 Avg Temp**) – The average temperature of the region in the year 2022, which influences the virus's transmission dynamics.
4. Cluster Mean CFR (**cluster_mean_CFR**) – The average case fatality rate within a defined cluster, providing insights into regional mortality risks.
5. Spatial Lag CFR (**spatial_lag_CFR**) – Captures the average case fatality rate of neighboring regions within a specified geographic radius, allowing the model to account for spatial dependencies in CFR.
6. Spatial Density (**spatial_density**) – Represents the density in a given area, which can affect the spread and fatality of the disease.
7. Spatial Lag (**spatial_lag**) – Captures the average value of deaths \ from neighboring regions, helping to model spatial influences on disease transmission.

Best Model

Model	RMSE
RandomForest	0.597
XGBoost	0.617
GradientBoosting	0.581
LinearRegression	0.596
Lasso	0.610
ElasticNet	0.604
Ridge	0.596
KNeighbors	0.604
Bagging_DecisionTree	0.593
Bagging_XGBoost	0.582
Bagging_KNeighbors	0.588
AdaBoost	0.757
SVM	0.623

Best Model: GradientBoosting

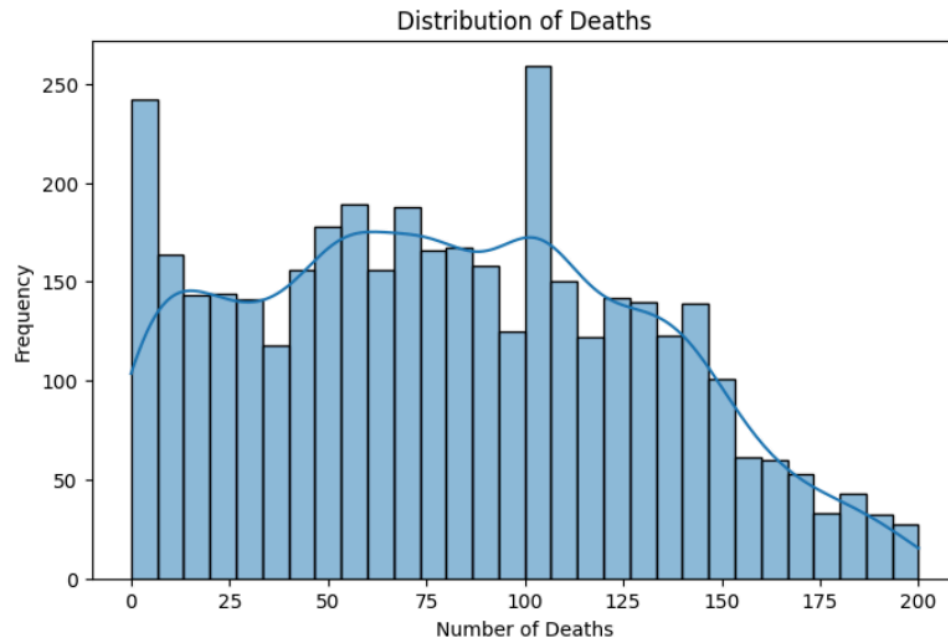
Best Feature Set: ('cluster_mean_CFR', 'spatial_lag_CFR', 'spatial_density')

Best RMSE: 0.57

Final Model RMSE: 0.57

After analyzing our results, we see that machine learning model outperform the deep learning models for death predictions, possibly due to the fact that the dataset we have is on the smaller side thus, we have decided to move forward with ML models.

2.Ball Tree Imputation-



Death Model:

Model	RMSE
RandomForest	33.29
XGBoost	35.54
GradientBoosting	32.56
LinearRegression	33.88
Lasso	33.87
ElasticNet	33.92
Ridge	33.88
KNeighbors	35.66
Bagging_DecisionTree	33.26
Bagging_XGBoost	32.69
Bagging_KNeighbors	35.00
AdaBoost	35.18
SVM	32.60

Best Model: Bagging_XGBoost

Best Feature Set: ('Lat', 'Long_', '2022 Avg Temp', 'cluster_mean_deaths')

Best RMSE: 32.41

Final Model RMSE: 32.41

CFR Model

Model	RMSE
RandomForest	0.594
XGBoost	0.605
GradientBoosting	0.579
LinearRegression	0.596
Lasso	0.610
ElasticNet	0.604
Ridge	0.596
KNeighbors	0.605
Bagging_DecisionTree	0.592
Bagging_XGBoost	0.580
Bagging_KNeighbors	0.590
AdaBoost	0.623
SVM	0.626

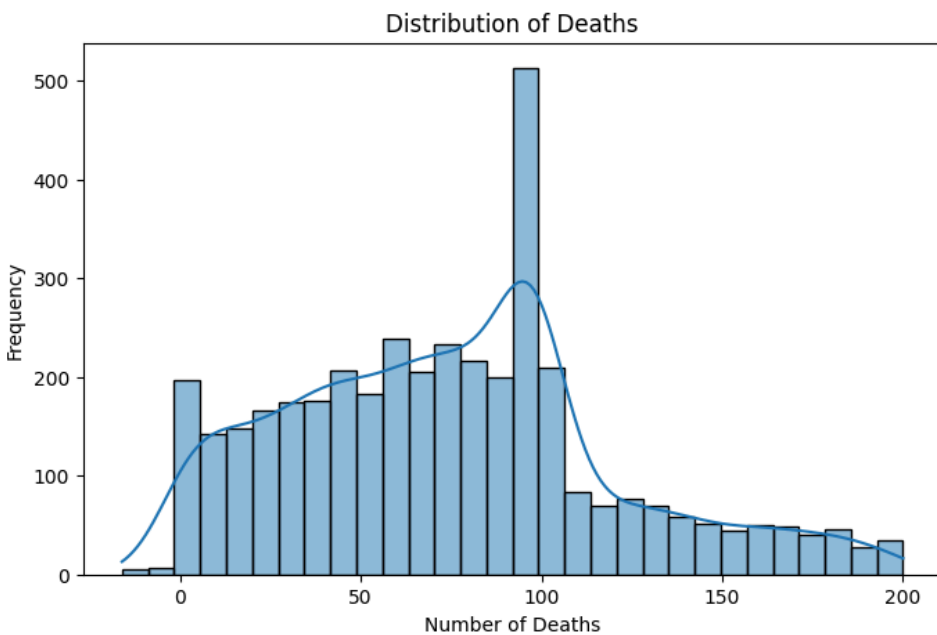
Best Model: GradientBoosting

Best Feature Set: ('cluster_mean_CFR', 'spatial_lag_CFR', 'spatial_density')

Best RMSE: 0.57

Final Model RMSE: 0.579

3.Kriging Imputation-



Death Model

Model	RMSE
RandomForest	33.93
XGBoost	36.18
GradientBoosting	33.62
LinearRegression	34.43
Lasso	34.43
ElasticNet	34.48
Ridge	34.43
KNeighbors	35.58
Bagging_DecisionTree	34.20
Bagging_XGBoost	33.51
Bagging_KNeighbors	34.70
AdaBoost	35.10
SVM	34.17

Best Model: Bagging_XGBoost

Best Feature Set: ('Lat', 'Long_', '2022 Avg Temp', 'is_tropical', 'cluster_mean_deaths')

Best RMSE: 32.88

Final Model RMSE: 32.88

CFR Model-

Model	RMSE
RandomForest	0.60
XGBoost	0.59
GradientBoosting	0.57
LinearRegression	0.60
Lasso	0.61
ElasticNet	0.60
Ridge	0.60
KNeighbors	0.60
Bagging_DecisionTree	0.60
Bagging_XGBoost	0.58
Bagging_KNeighbors	0.59
AdaBoost	0.62
SVM	0.63

Best Model: GradientBoosting
Best Feature Set: ('cluster_mean_CFR', 'spatial_lag_CFR', 'spatial_density')
Best RMSE: 0.57
Final Model RMSE: 0.57

Advanced NN with Residual Block-



Model Evaluation Results:

CFR Prediction RMSE: 0.5

9).Conclusion:

Death Model

Imputation	Best Model	Best Combination of Features	Best model RMSE (0.2/0.1 Train_test split)
IDW	Bagging_XGBoost	Lat, long_,2022 Avg Temp, Cluster Mean Deaths	32.23/31.14
Ball Tree	Bagging_XGBoost	Lat, Long_', '2022 Avg Temp', 'cluster_mean_deaths'	32.41
Kriging	Bagging_XGBoost	'Lat', 'Long_', '2022 Avg Temp', 'is_tropical', 'cluster_mean_deaths'	32.88

(train_test_split is 0.2/0.1 for all the experiments)

CFR Model

Imputation	Best Model	Best Combination Features	Best model Rsme (0.2/0.1 Train_test split)
IDW	Gradient Boosting	'cluster_mean_CFR', 'spatial_lag_CFR', 'spatial_density'	0.570/0.49
Ball Tree	Gradient Boosting	'Lat' 'cluster_mean_CFR', 'spatial_lag_CFR', 'spatial_density'	0.579
Kriging	Gradient Boosting	'cluster_mean_CFR', 'spatial_lag_CFR', 'spatial_density'	0.572

Model Postulate:

The **IDW and Bagging-XGBoost models**, combined with temperature and spatial features like spatial lag and cluster mean deaths, provide accurate predictions by capturing both local spatial dependencies and regional patterns. These models enhance prediction performance for geospatial data, especially when considering environmental and proximity-based influences.

The ensemble approach of Bagging-XGBoost enhances the model's ability to **generalize** by minimizing overfitting and improving accuracy. This is particularly useful when dealing with spatial features like **2022 average temperature** and **cluster mean deaths**, where interactions between variables are complex and non-linear. The final result is a model that can capture both **local trends** and **global patterns** effectively.

Gradient Boosting Model (GBM)

- **Objective:** Sequentially fits weak learners (typically decision trees) on residuals of previous models to minimize prediction error.
- **Key Features:**
 - Boosting technique
 - Reduces bias by focusing on difficult-to-predict instances.
 - Combines multiple weak models into a stronger model.
 - Can overfit if not tuned properly.
- **Hyperparameters:** Learning rate, number of estimators, max depth of trees, etc.

Bagging XGBoost

- **Objective:** Combines multiple XGBoost models trained on different subsets of data (bootstrapping) to reduce variance and overfitting.
- **Key Features:**
 - Bagging technique
 - Each model is trained on a different random subset of the data.
 - Reduces variance and improves stability.
 - Often leads to better generalization compared to a single model.
- **Hyperparameters:** Number of estimators, max depth, learning rate, and subsample fraction.

Hence, for the final prediction of Deaths and CFR, we will use IDW imputation followed by Bagging XGBoost for predicting Deaths with the given features. For CFR, we will use Gradient Boosting with its best features.